Topic 2.3 - Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________________ Class: ____________ Date: ____________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Analyze the structure, domain, range, monotonicity, concavity, and end behavior of exponential functions.
- Connect equal-interval proportional change to the parameters of an exponential formula.
- Describe and construct transformed exponential functions using precise asymptotic language.

Section A: Foundation and representation

Question 1

Core skill | No calculator

For $f(x) = 3(2)^x$, state the domain, range, y-intercept, horizontal asymptote, monotonicity, concavity, and both end behaviors.

Question 2

Core skill | No calculator

For $g(x) = -4(1/3)^{x-2} + 5$, state the horizontal asymptote, anchor point, range, and whether the function is increasing or decreasing.

Question 3

Core skill | No calculator

Find the y-intercept and exact x-intercept of $h(x) = 6(2)^{x+1} - 9$.

Question 4

Core skill | No calculator

An exponential function has asymptote $y = -3$, range $(-3, \infty)$, and anchor point $(2, 5)$. Write one possible function with base 4.

Question 5

Core skill | No calculator

Construct $f(x) = ab^x + 2$ through $(0, 8)$ and $(2, 56)$.

Question 6

Structure | No calculator

Construct an exponential function with asymptote $y = 10$ that passes through $(1, 6)$ and $(3, 9)$. Explain why the base represents decay even though the function increases.

Section B: Application, comparison, and error analysis

Question 7

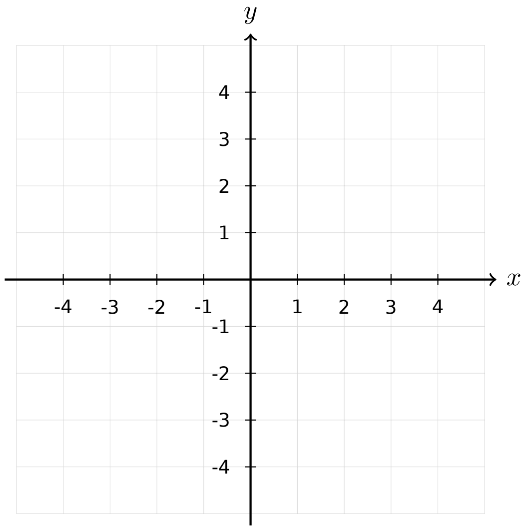
Core skill | No calculator

Order the functions 2^x, 3^x, and $(1.4)^x$ from least to greatest for (a) $x > 0$ and (b) $x < 0$.

Question 8

Graphing | No calculator

Sketch $p(x) = 2(1/2)^{x-1} - 4$. Label the asymptote, anchor point, and intercepts.



Question 9

Error analysis | No calculator

A student states that the range of $q(x) = -5(3)^x + 7$ is $(-\infty, \infty)$ because exponentials have all real outputs. Correct the statement.

Question 10

Reasoning | No calculator

Determine whether $f(x) = 4(0.7)^x - 1$ ever equals its horizontal asymptote. Justify algebraically.

Question 11

Modeling | No calculator

The function $T(t) = 18 + 62(0.76)^t$ models a temperature in degrees Celsius. Interpret 18, 62, and 0.76, and describe the long-run behavior.

Question 12

Parameter reasoning | No calculator

Find all values of k for which $f(x) = k(2)^x + 5$ is increasing and has range $(5, \infty)$.

Section C: AP-style multiple choice

MCQ 1

No calculator

Which function has horizontal asymptote $y = -2$ and range $(-2, \infty)$?

A. $-3(2)^x - 2$

B. $3(2)^x - 2$

C. $3(2)^x + 2$

D. $-3(1/2)^x + 2$

MCQ 2

No calculator

For $f(x) = a(0.4)^x + 7$, which condition makes f increasing?

A. $a > 0$

B. $a < 0$

C. $a = 0$

D. No value of a

MCQ 3

No calculator

Which point must lie on $g(x) = 5(3)^{x-4} - 1$?

A. $(4, 4)$

B. $(4, 5)$

C. $(-4, 4)$

D. $(0, 4)$

MCQ 4

No calculator

An exponential graph approaches $y = 6$ from below as $x \to -\infty$ and falls without bound as $x \to \infty$. Which structure is consistent?

A. $a > 0, b > 1, k = 6$

B. $a < 0, b > 1, k = 6$

C. $a > 0, 0 < b < 1, k = 6$

D. $a < 0, 0 < b < 1, k = 0$

Section D: Free-response reasoning

FRQ 1 - Full feature analysis

No calculator **6 points**

Let $f(x) = -6(2)^{x-3} + 10$. (a) State the domain, range, horizontal asymptote, and anchor point. (b) Find the y-intercept and the exact x-intercept. (c) Describe monotonicity, concavity, and end behavior. (d) Explain why the graph cannot cross its asymptote.

FRQ 2 - Construct from transformed data

No calculator **6 points**

An exponential function g has horizontal asymptote $y = 4$ and passes through $(1, 10)$ and $(4, 166)$. (a) Determine $g(x)$ in the form $ab^x + 4$. (b) Find $g(0)$ and interpret the point relative to the asymptote. (c) Solve $g(x) = 58$. (d) State the range and describe the effect of replacing a by $-a$.